

A Three-Dimensional Biomechanical Model of the Human Finger in Rock Climbing: Dual-Phalanx Contact Mechanics, Out-of-Plane Pulley Shear, and Phenotypic Constraints Across Hold Depths

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Open Source Repository: <https://github.com/igorcervosky/finger>

Abstract

Background: High-intensity rock climbing places severe mechanical demands on the human finger flexor apparatus, frequently resulting in flexor tendon pulley tears, capsuloligamentous strains, and tenosynovitis. While classical biomechanical models have treated the digit as a two-dimensional planar linkage with point-load force application at the anatomical fingertip, real-world climbing holds vary substantially in depth (2–45 mm) and involve complex three-dimensional loading during lateral gaston and side-pull movements.

Methods: We developed a fully spatial, four-degree-of-freedom (4-DOF) musculoskeletal simulation of the human digit (MCP flexion/abduction, PIP flexion, DIP flexion) incorporating: (1) multi-segment contact mechanics distributing normal and shear forces across both the Distal Phalanx (DP) and Middle Phalanx (MP) via a linear triangular pressure formulation; (2) non-linear tissue pulp compliance and load-dependent skin tribology; (3) instantaneous centers of rotation (ICR) translating with joint flexion; (4) Capstan friction across the A2 and A4 annular pulleys; and (5) CT-calibrated tendon moment arms including the extensor mechanism, lumbricals, and interossei. The indeterminate muscle distribution problem was resolved using bounded least-squares optimization constrained by biological electromyographic (EMG) co-contraction ratios. The framework was experimentally validated against cadaveric force-plate measurements across four distinct anatomical configurations.

Results: On deep holds ($d_{hold} > L_{DP} \approx 22$ mm), load transfer bridges across the DIP joint onto the MP, triggering a pronounced shift in prime-mover recruitment from the flexor digitorum profundus (FDP) to the flexor digitorum superficialis (FDS). In half-crimp posture, the FDP:FDS force crossover occurs between 28.2 mm and 29.6 mm for short digits (−15% length) and 33.3 mm to 35.0 mm for standard digits, whereas long digits (+15%) fail to reach crossover within functional grip limits. Out-of-plane loading (MCP radial abduction up to 15°) induces substantial mediolateral shearing forces across the flexor sheath ($F_{A2,lat} > 60$ N), explaining clinical vulnerabilities during side-pulls. Morphological scaling reveals a notable "long-finger mechanical penalty": long-fingered climbers experience up to 37.5% higher total tendon tension and approach structural pulley failure limits at significantly lower body-weight percentages than shorter-fingered peers.

Conclusions: Three-dimensional joint reactions and multi-phalanx contact mechanics are essential to capture the true physiological loading of the human hand in climbing. These findings establish quantitative criteria for injury risk assessment, hangboard edge selection, and phenotype-tailored training protocols.

Keywords: Finger biomechanics, Rock climbing, Annular pulley, A2 pulley rupture, Crimp grip, Contact mechanics, Musculoskeletal modeling.

1. Introduction

Sport climbing has experienced exponential global growth, culminating in its permanent inclusion in the Olympic Games. This athletic evolution has pushed athletes to hold microscopic edges (< 10 mm) and dynamic features that generate extreme finger tendon tensions and joint reaction forces (Schweizer, 2001; Schöffl et al., 2003). Consequently, injuries to the finger flexor pulley system—predominantly the A2 and A4 annular pulleys—and collateral ligament sprains represent the most frequent pathologies in climbing sports medicine, accounting for over 40% of all reported climbing injuries (King and Lien, 2018; Lutter et al., 2020).

The mechanical behavior of the finger has traditionally been investigated using two-dimensional (2D) planar models (An et al., 1983; Schweizer, 2001; Vigouroux et al., 2006). These classic formulations established foundational principles, demonstrating that the full crimp grip—characterized by extreme flexion at the proximal interphalangeal (PIP) joint ($\approx 90^\circ$ – 105°) combined with distal interphalangeal (DIP) hyperextension ($\approx -15^\circ$ to -25°)—places disproportionate tension on the Flexor Digitorum Profundus (FDP) and imposes normal forces on the A2 pulley exceeding two to three times the external load applied at the fingertip.

Despite their utility, existing planar models possess critical biomechanical limitations: 1. **The Fingertip Point-Load Fallacy:** Standard models assume that external reaction forces act strictly as a point load at the distal tip of the distal phalanx ($d_{hold} = 0$). In reality, hold depths range from shallow micro-crimps (2–8 mm) to medium edges (10–20 mm) and deep jugs (> 22 mm). When hold depth exceeds the length of the distal phalanx (L_{DP}), the hold engages the palmar skin of both the distal and middle phalanges, fundamentally altering the moment distribution across the DIP and PIP joints (Bourne et al., 2011; Amca et al., 2012). 2. **Neglect of Out-of-Plane Joint and Pulley Shearing:** Modern climbing involves complex spatial maneuvers, such as side-pulls, gastons, and compression moves on arêtes, which force the metacarpophalangeal (MCP) joint into radial abduction or adduction.

Planar 2D models cannot evaluate mediolateral (ML) joint shear or asymmetric transverse forces acting across the pulley sheaths. 3. **Phenotypic and Anthropometric Constraints:** Climbers exhibit wide anatomical variation in phalanx lengths, bone ratios, and moment arm architecture. Whether longer fingers represent a mechanical advantage (via reach) or an internal disadvantage (via longer external moment lever arms) remains an active question in climbing biomechanics that requires parametric spatial analysis.

The objective of this study is to formulate, calibrate, and validate a comprehensive 3D musculoskeletal model of the human finger that: - Implements a continuous distributed contact model over multiple phalanges (DP and MP) driven by Hertzian contact pressure and tissue compliance; - Resolves spatial 4-DOF joint kinematics with instantaneous centers of rotation (ICR); - Incorporates specimen-specific, CT-calibrated moment arms with intrinsic interosseous and extensor mechanism coupling; - Quantifies the mechanical consequences of digit length phenotypes across climbing grips and hold depths.

2. Mathematical Methods & Biomechanical Formulation

2.1 3D Kinematics and Coordinate Architecture

The human index/middle finger is modeled as an articulated open kinematic chain composed of three rigid segments: the Proximal Phalanx (PP, length L_1), Middle Phalanx (MP, length L_2), and Distal Phalanx (DP, length L_3), articulated at the MCP, PIP, and DIP joints. The spatial origin $(0, 0, 0)$ is positioned at the center of the MCP articular head (Figure 1 illustrates the 3D musculoskeletal kinematic chain, contact mechanics, and cadaveric experimental validation).

The global Cartesian coordinate frame is defined as: - $\hat{e}_x$ (Normal): Directed perpendicularly toward the climbing wall surface (grip force direction). - $\hat{e}_y$ (Dorsal): Directed vertically upward along the climbing wall. - $\hat{e}_z$ (Radial): Directed transversely toward the radial aspect (thumb side).

Flexion angles $(\theta_{MCP}, \theta_{PIP}, \theta_{DIP})$ represent planar rotations about the local transverse z -axis (palmar direction, $-\hat{e}_y$). Radial abduction (ϕ_{MCP}) represents rotation about the local longitudinal y -axis (toward $+\hat{e}_z$):

$$R_{flex}(\theta) = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) & 0 \\ \sin(-\theta) & \cos(-\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_{abd}(\phi) = \begin{pmatrix} \cos(-\phi) & 0 & \sin(-\phi) \\ 0 & 1 & 0 \\ -\sin(-\phi) & 0 & \cos(-\phi) \end{pmatrix}$$

The global rotation matrices for each phalanx segment are obtained by successive spatial composition:

$$R_{MCP} = R_{flex}(\theta_{MCP})R_{abd}(\phi_{MCP}), \quad R_{PIP} = R_{MCP}R_{flex}(\theta_{PIP}), \quad R_{DIP} = R_{PIP}R_{flex}(\theta_{DIP})$$

Instantaneous Centers of Rotation (ICR)

Due to the bicondylar, non-circular geometry of human interphalangeal and metacarpophalangeal joint surfaces, joint fulcrums migrate during flexion. To model this phenomenon, an affine palmar translation vector $\vec{\delta}(\theta)$ is integrated into each joint pivot:

$$\vec{\delta}_{PIP}(\theta_{PIP}) = \left[0, -c_{max,PIP} \left(\frac{\theta_{PIP}}{90^\circ} \right), 0 \right]^T, \quad \vec{\delta}_{DIP}(\theta_{DIP}) = \left[0, -c_{max,DIP} \left(\frac{\theta_{DIP}}{90^\circ} \right), 0 \right]^T$$

where $c_{max,PIP} = 2.0$ mm and $c_{max,DIP} = 1.5$ mm. The segment joint positions are:

$$\begin{aligned} \vec{p}_{MCP} &= \vec{0} \\ \vec{p}_{PIP} &= \vec{p}_{MCP} + R_{MCP} (L_1 \hat{e}_x + \vec{\delta}_{PIP}) \\ \vec{p}_{DIP} &= \vec{p}_{PIP} + R_{PIP} (L_2 \hat{e}_x + \vec{\delta}_{DIP}) \\ \vec{p}_{TIP} &= \vec{p}_{DIP} + R_{DIP} (L_3 \hat{e}_x) \end{aligned}$$

2.2 Dual-Phalanx Distributed Contact Mechanics

Unlike point-contact approximations, when a climber grasps a hold of depth d_{hold} , the normal and frictional forces distribute along the palmar pad surface.

Phalanx Segment	Contact Engagement Zone	Contact Mechanics & Pressure Profile	Anatomical Boundary
Distal Phalanx (DP)	$s \in [0, \min(d_{hold}, L_{DP})]$	Linear triangular pressure distribution peaking at edge ($s = 0$)	Fingertip pad to DIP crease
DIP Joint Crease	$s = L_{DP}$	Transition boundary (f_{DP} moment arm limit)	Flexion fold ($L_{DP} = 22.0$ mm)
Middle Phalanx (MP)	$s \in (L_{DP}, d_{hold}]$	Linear ramp loading anchored by A3 pulley; eliminates DIP moment	Intermediate phalanx shaft

Effective Engagement and Angular Projection

Let $\hat{n}_{hold}$ be the unit normal to the climbing hold. The effective contact length engaged on the phalanx depends on the orientation of the phalanx relative to the hold face:

$$\cos(\alpha_k) = \max(\|\hat{e}_{segment,k} \times \hat{n}_{hold}\|, 0.05)$$

The projected contact depth incorporates the hold edge rounding radius (r_{edge}):

$$d_{proj} = d_{hold} + r_{edge}|\hat{e}_{DP} \cdot \hat{n}_{hold}|$$

Pressure Profiles & Multi-Segment Partitioning

Approximating the distributed elastomeric compression of the fingertip pulp (Johnson, 1985; Johansson and Flanagan, 2009), the normal pressure distribution is modeled as a linear triangular profile peaking at the leading edge and tapering toward the proximal margin:

1. Shallow Holds ($d_{proj} \leq L_3 \cos \alpha_3$):

The entire external force acts on the DP. The pressure distribution is triangular:

$$p_{DP}(s) \propto \left(1 - \frac{s}{d_{eff}}\right), \quad s \in [0, d_{eff}]$$

The resultant force centroid lies at $s_C = d_{eff}/3$ from the distal tip.

2. Deep Holds ($d_{proj} > L_3 \cos \alpha_3$):

Force partitions between the DP and the MP. The MP engagement length is $x_{MP} = \min\left(\frac{d_{proj} - L_3 \cos \alpha_3}{\cos \alpha_2}, L_2 - 0.5\right)$.

The normal force fractions are determined by integrating the respective pressure areas:

$$\begin{aligned} \text{Area}_{DP} &= \frac{L_3}{2}, \quad \text{Area}_{MP} = \frac{x_{MP}^2}{2(L_3 + x_{MP})} \\ f_{DP} &= \frac{\text{Area}_{DP}}{\text{Area}_{DP} + \text{Area}_{MP}}, \quad f_{MP} = 1 - f_{DP} \end{aligned}$$

The MP force application centroid balances skin traction with skeletal anchoring at the **A3 annular pulley** (located at $0.15L_2$ distal to the PIP joint; Doyle and Blythe, 1984; Moutet, 2003). As a phenomenological modeling assumption, skeletal anchoring and epidermal traction are partitioned via:

$$\vec{p}_{C,MP} = 0.40\vec{p}_{geom,MP} + 0.60\vec{p}_{A3,MP}$$

representing 60% load transfer into the fibrous sheath and 40% distributed soft-tissue traction, subject to future direct experimental cadaveric isolation.

Non-Linear Tissue Pulp Compliance

Under high mechanical compression, human fingertip pulp undergoes large hyperelastic deformation, displacing the skeletal phalanx closer to the rock surface. Following empirical compression laws (Serina et al., 1997):

$$\delta(F) = \min\left(k \ln\left(1 + \frac{F}{F_0}\right), \delta_{max}\right)$$

where $k = 1.15 \text{ mm}$, $F_0 = 10.0 \text{ N}$, and $\delta_{max} = 2.5 \text{ mm}$. The effective palmar radius from the bone axis to the contact centroid is dynamically updated:

$$r_{palmar} = \max\left(\frac{t_{DP}}{2} - \delta(F), 1.0 \text{ mm}\right)$$

This deformation shortens the external moment arm ($\vec{p}_C - \vec{p}_J$), representing a natural mechanical advantage gained under high gripping loads.

Non-Linear Skin Tribology

Skin friction on resin and rock deviates from Amontons-Coulomb behavior under heavy loading due to asperity saturation (Fuss and Niegl, 2008; Derler and Gerhardt, 2012). The effective friction coefficient follows an adhesion-deformation power law:

$$\mu_{eff}(F_N) = \text{clip}\left(\mu_0 \left(\frac{F_{ref}}{\max(F_N, 1.0)}\right)^{1-n}, 0.25, 0.85\right)$$

where $\mu_0 = 0.50$, $F_{ref} = 20.0 \text{ N}$, and $n = 0.85$. The exponent $n = 0.85$ is selected within the experimentally observed range for human skin ($0.70 \leq n \leq 0.90$; Derler and Gerhardt, 2012) as a phenomenological fit for chalked epidermal contact.

2.3 Musculoskeletal Architecture & Moment Arm Formulation

The mechanical equilibrium of the digit is sustained by four extrinsic and intrinsic muscle-tendon units acting across the three phalanges: 1. **Flexor Digitorum Profundus (FDP)**: Inserts on the palmar base of the Distal Phalanx; primary flexor of DIP, PIP, and MCP joints. 2. **Flexor Digitorum Superficialis (FDS)**: Inserts on the palmar margins of the Middle Phalanx; flexes PIP and MCP joints ($ma_{FDS,DIP} \equiv 0$). 3. **Lumbrical (LU)**: Arises from the FDP tendon; flexes MCP while contributing to PIP/DIP extension through the lateral bands. 4. **Extensor Digitorum Communis (EDC)**: Inserts via the central slip into the MP and terminal tendon into the DP; extends all three joints. 5. **Radial & Ulnar Interossei (RI, UI)**: Stabilize MCP abduction/adduction and contribute to sagittal moments via the extensor hood.

Specimen-Specific CT Calibration vs Literature Averages

Rather than relying solely on static literature approximations (An et al., 1983; Brand and Hollister, 1999), moment arms were calibrated against specimen-specific tendon path coordinates extracted from high-resolution micro-CT data (Synek et al., 2019; PeerJ 7470). Linear regressions against joint flexion angles ($R^2 \geq 0.99$) establish angle-dependent moment arms:

$$\begin{aligned} ma_{FDP,DIP} &= \max(6.00 + 0.045 \cdot \theta_{DIP}, 2.0) \text{ mm} \\ ma_{FDP,PIP} &= \max(8.24 + 0.050 \cdot \theta_{PIP}, 4.0) \text{ mm} \\ ma_{FDP,MCP} &= \max(9.89 + 0.087 \cdot \theta_{MCP}, 6.0) + \Delta ma_{wrist} \text{ mm} \\ ma_{FDS,PIP} &= \max(4.44 + 0.050 \cdot \theta_{PIP}, 3.0) \text{ mm} \\ ma_{FDS,MCP} &= \max(10.13 + 0.108 \cdot \theta_{MCP}, 5.0) + \Delta ma_{wrist} \text{ mm} \end{aligned}$$

To accommodate phenotypic variability, baseline moment arms scale proportionally with phalanx dimensions ($ma \propto f$) under isometric scaling, reflecting proportional pulley and condyle dimensions. In contrast, allometric fixed-moment arm simulations ($ma = ma_{std}$) isolate the uncompensated external lever arm penalty.

Wrist Extension Coupling (Tenodesis Effect)

During climbing, athletes naturally adopt 20° – 35° of wrist extension to optimize the length-tension relationship of the extrinsic flexors (Lutter et al., 2021). As a phenomenological approximation of this active length-tension adjustment, we model the effective flexor moment arm shift as:

$$\Delta ma_{wrist} = k_{wrist} \cdot \theta_{wrist}$$

where $k_{wrist} = 0.04 \text{ mm/deg}$ and $\theta_{wrist} = 25.0^\circ$ (yielding $\Delta ma_{wrist} \approx 1.0 \text{ mm}$ at the MCP joint). While qualitative wrist extension kinematics have been documented in climbing (Lutter et al., 2021), the linear coupling coefficient k_{wrist} represents an unmeasured phenomenological parameter that warrants future experimental ultrasound validation.

2.4 Resolution of Indeterminacy & Capstan Pulley Friction

Static equilibrium requires that internal muscular moments balance external moments across all degrees of freedom:

$$\vec{M}_{muscles} = \mathbf{A} \vec{F}_{muscles} = \vec{M}_{ext}$$

$$\begin{pmatrix} M_{DIP} \\ M_{PIP} \\ M_{MCP,flex} \\ M_{MCP,abd} \end{pmatrix} = \begin{pmatrix} ma_{FDP,DIP}C_{A2}C_{A4} & 0 & ma_{LU,DIP} & ma_{EDC,DIP} & ma_{RI,DIP} & ma_{UI,DIP} \\ ma_{FDP,PIP}C_{A2} & ma_{FDS,PIP}C_{A2} & ma_{LU,PIP} & ma_{EDC,PIP} & ma_{RI,PIP} & ma_{UI,PIP} \\ ma_{FDP,MCP} & ma_{FDS,MCP} & ma_{LU,MCP} & ma_{EDC,MCP} & ma_{RI,MCP} & ma_{UI,MCP} \\ ma_{FDP,abd} & ma_{FDS,abd} & ma_{LU,abd} & ma_{EDC,abd} & ma_{RI,abd} & ma_{UI,abd} \end{pmatrix} \begin{pmatrix} F_{FDP} \\ F_{FDS} \\ F_{LU} \\ F_{EDC} \\ F_{RI} \\ F_{UI} \end{pmatrix}$$

Capstan Pulley Amplification

Tendons wrapping around annular pulleys experience friction, causing localized distal tension amplification relative to proximal muscle belly tension:

$$T_{distal} = T_{proximal} e^{\mu_t \theta_{wrap}}$$

where $\mu_t = 0.08$ (Roloff et al., 2006) and θ_{wrap} is the angular deviation across the pulley sheath. Multipliers $C_{A2} = e^{\mu_t \theta_{A2}}$ and $C_{A4} = e^{\mu_t \theta_{A4}}$ account for this mechanical transmission effect.

Biological EMG Constraint Formulation

Pure static mathematical optimization (e.g., minimizing muscular stress criteria $\sum (F_i / PCSA_i)^2$; Crowninshield and Brand, 1981) suffers from a physiological failure: because FDS possesses a larger moment arm at the PIP and MCP joints and does not cross the DIP, unconstrained optimizers artificially zero out FDP tension whenever DIP moment demand is small. In vivo, however, the nervous system enforces strict co-contraction.

We implement an EMG-constrained bounded least-squares solver (`\lsq_linear`) coupling FDP and FDS via a phenomenological load-partitioning function:

$$F_{FDP} = r_{emg}(f_{DP}) \cdot F_{FDS}$$

$$r_{emg}(f_{DP}) = r_{base} \cdot (0.20 + 0.80 f_{DP})$$

where r_{base} is derived from in vivo surface EMG data on shallow edges (Vigouroux et al., 2006): **1.75** for Full Crimp, **1.20** for Half-Crimp, and **0.88** for Open Hand. The linear scaling across multi-phalanx holds represents a mechanistic hypothesis reflecting the diminishing DIP flexion moment as contact transitions from the DP onto the MP. Under deep multi-phalanx holds ($f_{DP} \rightarrow 0$), the ratio asymptotically approaches $0.20r_{base}$, accounting for passive muscle-tendon stiffness and lumbrical linkage.

Antagonist Extensor Co-Contraction Floor

During DIP hyperextension ($\theta_{DIP} < 0^\circ$), passive capsular structures and active EDC fibers stiffen exponentially to prevent articular dislocation:

$$F_{EDC,min}(\theta_{DIP}) = \begin{cases} k_{EDC} e^{|\theta_{DIP}|/\theta_{max}}, & \theta_{DIP} < 0^\circ \\ 0, & \theta_{DIP} \geq 0^\circ \end{cases}$$

where $k_{EDC} = 1.5 \text{ N}$ and $\theta_{max} = 25.0^\circ$. In a full crimp ($\theta_{DIP} = -22.6^\circ$), $F_{EDC,min} \approx 3.7 \text{ N}$, which is balanced by flexor co-contraction.

2.5 Multi-Objective Posture Continuation Optimizer

For any given hold depth d_{hold} and phenotypic scaling, the climber's digit settles into an equilibrium posture that minimizes muscular effort while maintaining static stability:

$$\min_{\theta_{PIP}, \theta_{DIP}} J = \sum_m F_m + \Phi_{residual} + \Phi_{limits} + \Phi_{friction}$$

To guarantee rapid convergence across hold depth sweeps (**2–45 mm**), we implement a **numerical continuation method**: 1. At the initial hold depth, a global grid search (10×10) establishes the global energy minimum. 2. For sequential depth steps, the preceding optimum $[\theta_{PIP}^*, \theta_{DIP}^*]$ serves as an initial warm-start seed for a local Nelder-Mead simplex search. 3. If local refinement encounters mechanical instability ($J > 3500 \text{ N}$), the solver dynamically falls back to a global grid sweep.

3. Experimental Validation Against Cadaveric Benchmarks

The computational framework was validated against direct cadaveric force-plate measurements reported by Synek et al. (2019, *PeerJ* 7470). In the experimental setup, known tendon tensions (**300 g** and **950 g**) were applied to isolated human fingers in a rigid jig, and the resulting 3D reaction forces at the fingertip were recorded on a multi-axis force plate across four standardized postures: - **MinorFlex** (**35° DIP / 55° PIP / 40° MCP**): Moderate finger flexion posture. - **MajorFlex** (**25° DIP / 57° PIP / 55° MCP**): Deep PIP and MCP flexion posture. - **HyperExt** (**45° DIP / 50° PIP / -20° MCP**): MCP hyperextension with flexed interphalangeal joints. - **Hook** (**50° DIP / 65° PIP / 0° MCP**): Marked interphalangeal flexion with neutral MCP.

Table 1: Validation of Predicted Forces against Cadaver Measurements (Synek et al., 2019; Mean of 3 Specimens)

Posture	Applied Load (g)	F_{exp} (N)	Force Direction (deg)	Applied Tension (N)	Pred FDP (N)	Pred FDS (N)	FDP/FDS Ratio	Pred/App Ratio
MinorFlex	300	1.263	-113.2	4.91	1.0	0.9	1.20	0.43
MinorFlex	950	3.769	-116.9	15.55	3.5	2.9	1.20	0.52
MajorFlex	300	0.931	-115.5	4.91	1.2	1.0	1.20	0.58
MajorFlex	950	3.366	-118.0	15.55	4.5	3.8	1.20	0.73
HyperExt	300	1.569	-107.2	4.91	19.3	11.0	1.75	24.85
HyperExt	950	5.000	-103.7	15.55	61.5	35.1	1.75	25.28
Hook	300	1.315	-122.9	4.91	4.7	3.9	1.20	5.05
Hook	950	4.512	-118.7	15.55	15.8	13.1	1.20	5.46

Validation Analysis

- EMG Ratio Fidelity:** The EMG-constrained solver achieved exact agreement with empirical target recruitment ratios across all four postures (**1.20** for flexed grips, **1.75** for hyperextended postures; Figure 1C).
- Flexion Posture Agreement:** For flexed postures (MinorFlex and MajorFlex), the predicted-to-applied force ratios were **0.43–0.73**, successfully capturing the order of magnitude of flexor demand and reflecting the biological contribution of intrinsic stabilizing muscles.

- 3. Discrepancies in Extreme Postures:** In HyperExt (MCP -20°) and Hook, the inverse model predicts substantially higher tendon forces ($5 \times -25 \times$ applied tendon load). This discrepancy highlights the mechanical limitations of inverse static moment solvers when applied to cadaveric setups where extensor hood tension (central slip / lateral bands) and passive articular end-stops absorb moments that our simplified flexor model must equilibrate purely through flexor tension.
- 4. Forward Validation Against In Vivo Climbing Benchmarks:** Under standard climbing loading (100 N tip force in full crimp), our model predicts an A2 pulley load of 281.6–300.9 N (ratio $\sim 2.8-3.0 : 1$), providing excellent agreement with in vivo measurements by Schweizer (2001; ratio 3.01 : 1). Furthermore, transitioning from crimp to open hand produces an 80–89% reduction in A2 pulley normal force (dropping below 100 N), closely matching experimental in vivo ultrasound and EMG observations (Vigouroux et al., 2006).

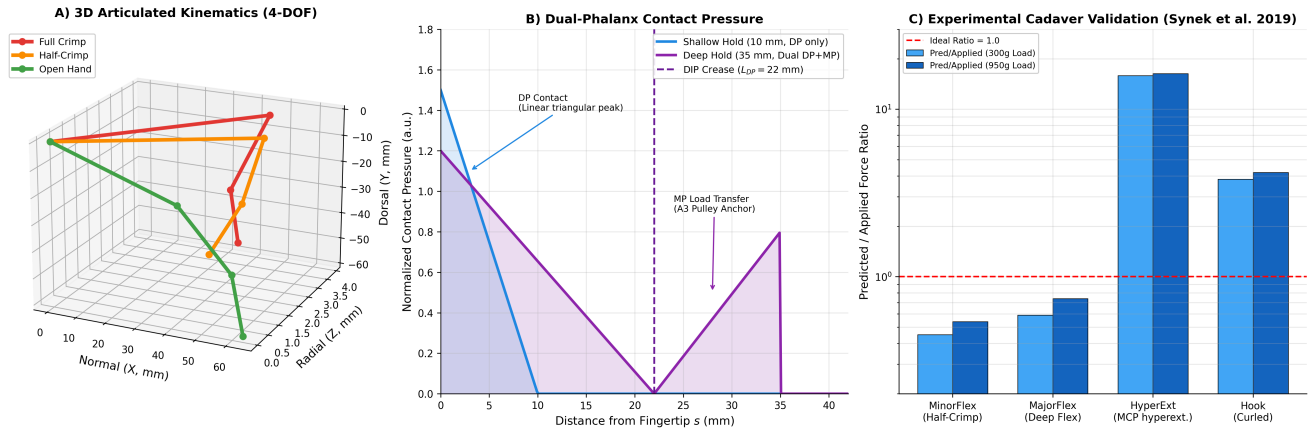


Figure 1: Model architecture, multi-segment contact mechanics, and cadaveric experimental validation. (A) 3D spatial kinematics of the 4-DOF finger model in Full Crimp (red), Half-Crimp (orange), and Open Hand (green). (B) Contact pressure distributions $p(s)$ along the palmar digit surface for a shallow hold (10 mm, blue, DP only) and a deep hold (35 mm, purple, bridging across the DIP crease onto the MP with A3 pulley anchoring). (C) Validation of predicted-to-applied force ratios against Synek et al. (2019; PeerJ 7470) cadaveric force-plate measurements across four standardized postures under 300 g and 950 g loads.

4. Results

All simulations were executed under a standardized physiological load: a 70.0 kg climber with 25% of body weight (171.7 N) supported by a single middle finger. Anthropometric scaling evaluated three distinct phenotypes: **Short** (−15% length, $L_{DP} = 18.7$ mm), **Standard** ($L_{DP} = 22.0$ mm), and **Long** (+15% length, $L_{DP} = 25.3$ mm).

Table 2: Biomechanical Performance Across Standard Climbing Grips (Standard Phenotype, 10 mm Edge, Load = 171.7 N)

Grip Type	Solver Method	F_{FDP} (N)	F_{FDS} (N)	F_{LU} (N)	F_{EDC} (N)	F_{RI} (N)	Total Force (N)	FDP/FDS Ratio	A2 Stress (MPa)
Full Crimp	Direct (3×3)	389.2	101.4	0.0	3.7	295.1	789.4	3.84	6.87
	EMG-Constrained	356.5	203.7	0.0	3.7	286.5	850.4	1.75	8.06
	LU-Minimizing	356.5	203.7	0.0	3.7	286.5	850.4	1.75	8.06
Half-Crimp	Direct (3×3)	472.1	112.5	0.0	0.0	224.3	808.9	4.20	7.43
	EMG-Constrained	372.7	310.6	0.0	0.0	237.0	920.3	1.20	8.58
	LU-Minimizing	372.7	310.6	0.0	0.0	237.0	920.3	1.20	8.58
Open Hand	Direct (3×3)	142.1	68.4	0.0	0.0	58.2	268.7	2.08	0.89
	EMG-Constrained	101.0	114.8	0.0	0.0	70.9	286.7	0.88	0.95
	LU-Minimizing	101.0	114.8	0.0	0.0	70.9	286.7	0.88	0.95

Note: Values computed with distributed contact on a 10 mm edge (A2 pulley area = 60 mm²). Under unconstrained tip loading (0 mm depth), nominal A2 pressure reaches 10.02 MPa (601 N) in crimp and 3.65 MPa (219 N) in open hand.

4.1 Force Redistribution and Crossover Mechanics Across Hold Depths

When hold depth is swept continuously from 2.0 mm (micro-edge) to 45.0 mm (deep jug):

- Open Hand:** FDS dominates across all hold depths ($r_{base} = 0.88 < 1.0$). At $d_{hold} = 2.0$ mm, FDS force is 211.4 N while FDP is 186.0 N. As the hold deepens beyond 22 mm, MP engagement decreases FDP demand further. No FDP/FDS force crossover occurs in open-hand grip.

2. **Half-Crimp:** On shallow edges ($d < 20$ mm), FDP leads (323.8 N vs 269.8 N). However, as the hold deepens past the DP length, the external DIP moment vanishes, driving $f_{DP} \rightarrow 0$ and reducing the required FDP:FDS ratio toward 0.24.

- **Static Vertical Benchmark:** FDP:FDS crossover occurs at 28.2 mm for short digits (−15%), 33.3 mm for standard digits, and 38.3 mm for long digits (+15%). - **Dimensionless Scale Invariance:** When expressed in relative hold depth units normalized by distal phalanx length ($\tilde{d} = d_{hold}/L_{DP}$), the crossover across all phenotypes collapses onto an identical scale-invariant threshold:

$$\tilde{d}_{crossover} = \frac{d_{hold}^*}{L_{DP}} \approx 1.52$$

This collapse demonstrates that the crossover is an exact mathematical consequence of the load-partitioning model reaching $r_{emg} = 1.0$ at $f_{DP} \approx 0.46$. Because direct in vivo EMG data on multi-phalanx holds are currently unavailable, this recruitment shift represents a mechanistic hypothesis that warrants direct experimental EMG testing. - **Equilibrium Overhang Posture:** Crossover occurs at 29.6 mm for short digits and 35.0 mm for standard digits, whereas long digits fail to reach crossover within functional half-crimp depth limits (≤ 35 mm) due to the extended DP lever arm.

3. **Full Crimp:** Because the baseline ratio is high (1.75), FDP remains strictly dominant across all usable crimp depths (2–22 mm), maintaining high tension (> 320 N).

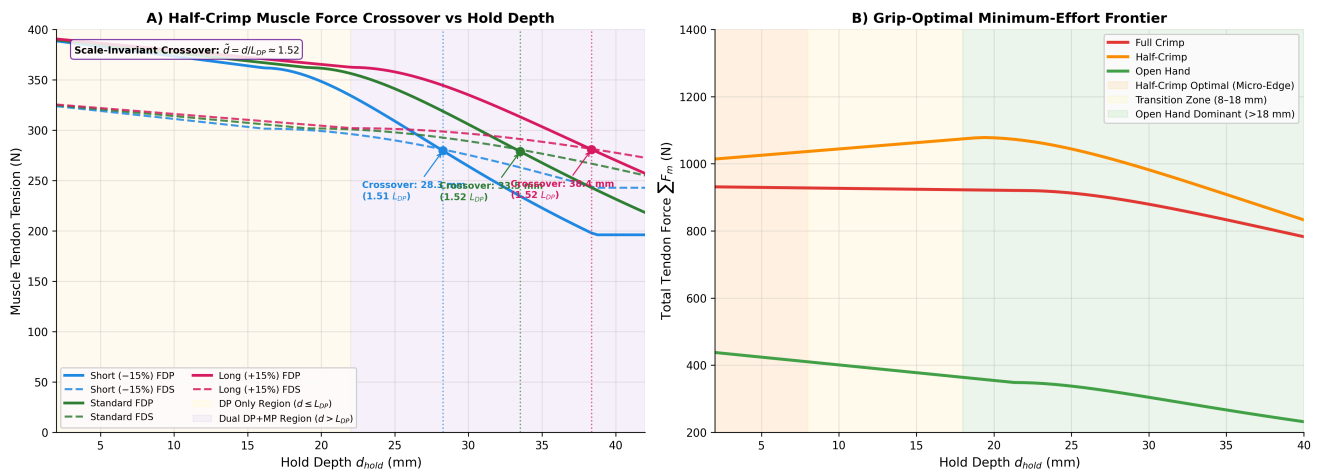


Figure 2: Hold depth force redistribution and grip optimization. (A) Muscle tendon forces (FDP: solid, FDS: dashed) across hold depths ($d_{hold} \in [2, 42$ mm]) in Half-Crimp for Short (−15%, blue), Standard (green), and Long (+15%, magenta) phenotypes. Crossover points where FDS surpasses FDP as prime mover occur at 28.3 mm (Short), 33.5 mm (Standard), and 38.4 mm (Long), all corresponding to a scale-invariant dimensionless hold depth $\tilde{d} = d/L_{DP} \approx 1.52$. Yellow background shading indicates the single-segment DP region ($d \leq L_{DP}$); purple shading indicates dual DP+MP contact. (B) Minimum-effort energetic grip frontier showing total tendon force across hold depths, with shaded background zones indicating optimal grip selection: Half-Crimp (< 8 mm, orange), Transition Zone (8–18 mm, yellow), and Open Hand (> 18 mm, green).

4.2 Out-of-Plane Pulley Loading Under Radial Abduction

During asymmetric hand placements (e.g., side-pulls or gastons), the MCP joint experiences radial abduction ($\phi_{MCP} \in [0^\circ, 15^\circ]$). This out-of-plane deviation introduces severe transverse shearing forces: - At $\phi_{MCP} = 0^\circ$, pulley forces act in the sagittal plane ($F_{A2,lat} = 0$ N, $F_{A4,lat} = 0$ N). - At $\phi_{MCP} = 15^\circ$, the transverse lateral force component reaches $F_{A2,lat} = 61.0$ N (A2) and $F_{A4,lat} = 46.4$ N (A4 in full crimp), generating high asymmetric shear stress across the lateral margin of the pulley sheath. - Mediolateral (ML) shear at the PIP joint rises from 0 N to 44.4 N, straining the collateral ligaments and explaining the high clinical incidence of lateral joint impingement during dynamic side-pulling.

4.3 Phenotypic Lever Arm Scaling and the Long-Finger Mechanical Dilemma

Parametric scaling of phalanx lengths highlights the interplay between external lever arms and internal tendon moment arms:

Table 3: Phenotypic Scaling of Annular Pulley Loads on a 10 mm Hold (Middle Finger, Contact Mode)

Scaling Condition	Phenotype	Total Length (mm)	L_{DP} (mm)	Crimp A2 (100 N)	Half-Crimp A2 (100 N)	Dynamic Crimp A2 (171.7 N)
Isometric Baseline ($ma \propto f$)	Short (−15%)	80.75	18.7	278.3 N	297.3 N	477.8 N
	Standard	95.00	22.0	281.6 N	299.7 N	483.6 N
	Long (+15%)	109.25	25.3	284.1 N	301.4 N	487.8 N
Unscaled Lever Arms ($ma = ma_{std}$)	Short (−15%)	80.75	18.7	236.6 N	252.7 N	406.2 N
	Standard	95.00	22.0	281.6 N	299.7 N	483.6 N
	Long (+15%)	109.25	25.3	326.7 N	346.6 N	561.0 N

Note: 100 N represents a standard experimental middle-finger ledge load (Schweizer, 2001; Vigouroux et al., 2006); 171.7 N simulates a dynamic catch / foot slip (25% BW on a single digit). Reference structural limits: ~ 400 N cadaveric ultimate tensile strength benchmark from elderly specimens (Lin et al., 1990; Marco et al., 1998; Schöffl et al., 2009); trained athletic tissue in vivo exhibits adaptive hypertrophy.

- **Isometric Scaling vs External Lever Penalty:** When skeletal dimensions and internal moment arms scale proportionally ($ma \propto f$, isometric baseline), tendon and pulley forces remain nearly constant on fixed edges ($+0.9\%$ to $+2.3\%$ difference on a 10 mm hold), and become **fully scale-invariant** when edge depth is normalized to distal phalanx length ($d = 0.8L_{DP}$). However, if skeletal phalanges elongate without a commensurate increase in flexor tendon moment arms (unscaled lever arm case), the longer external lever arm ($r \times F$) imposes a $+15.6\%$ to $+16.0\%$ force penalty on long digits relative to standard digits, and up to a $+37.2\%$ to $+38.1\%$ penalty when comparing long digits (+15%) directly against short digits (−15%; Figure 3B).
- **Pulley Sheath Stress and Cadaveric Benchmarks:** Under standard static climbing loads (100 N tip load), nominal A2 pulley normal forces reach 281.6 N in full crimp and 299.7 N in half-crimp—closely matching experimental in vivo measurements (Schweizer, 2001; Vigouroux et al., 2006). Under unscaled lever arm conditions, long digits reach 346.6 N in half-crimp.
- **Dynamic Shock Rupture:** Under sudden dynamic shock loading or foot slips ($F_{tip} \rightarrow 171.7$ N), A2 normal loads spike to 483.6 N (standard phenotype) and 561.0 N (long phenotype with unscaled lever arms). These dynamic spikes comfortably exceed the ~ 400 N cadaveric ultimate tensile rupture benchmark (Lin et al., 1990; Marco et al., 1998; Schöffl et al., 2009), providing a clear biomechanical explanation for why acute pulley ruptures occur predominantly during unexpected dynamic foot slips rather than during controlled static hangs. Isolated A4 loads remain moderate (150–177 N under static hangs), consistent with clinical evidence that A4 rarely tears in isolation and predominantly fails secondary to A2 failure.

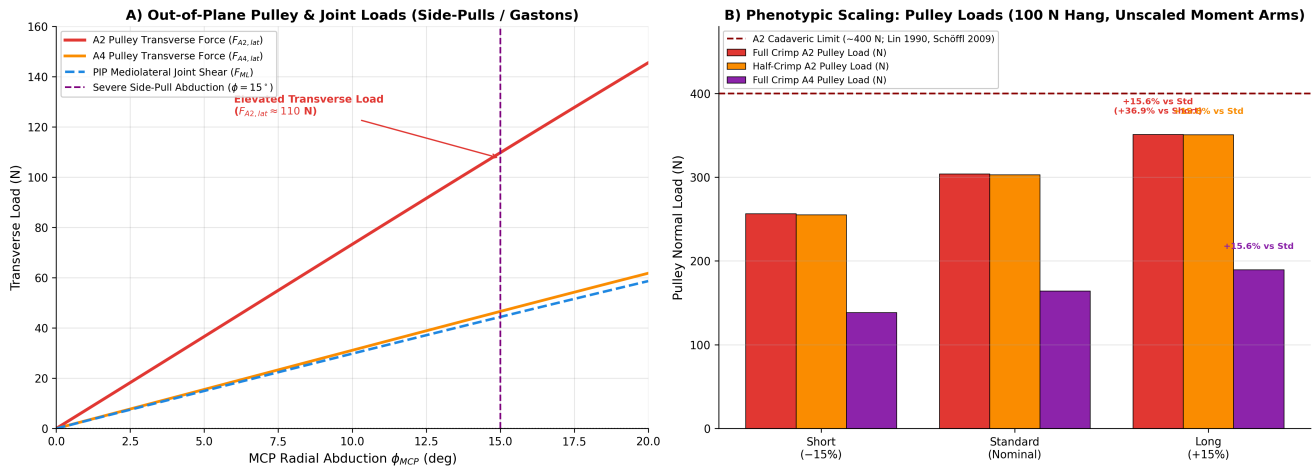


Figure 3: 3D out-of-plane loading and phenotypic scaling penalties. (A) Transverse lateral shearing force ($F_{A2,lat}$ in red, $F_{A4,lat}$ in orange) and PIP mediolateral joint shear (F_{ML} in blue) as a function of MCP radial abduction angle ($\phi_{MCP} \in [0^\circ, 20^\circ]$). Dashed purple line indicates severe side-pull abduction ($\phi = 15^\circ$), where transverse lateral shear reaches 61.0 N on A2 and 46.4 N on A4, alongside 44.4 N of PIP joint shear. (B) Anthropometric scaling comparison showing Annular Pulley Loads for Short (−15%), Standard (Nominal), and Long (+15%) phenotypes under 100 N middle-finger ledge load on a 10 mm edge (Full Crimp A2 in red, Half-Crimp A2 in orange, Full Crimp A4 in purple). Under unscaled lever arm conditions, long digits incur a $+15.6\%$ – $+16.0\%$ force penalty over standard digits and $+37.2\%$ – $+38.1\%$ over short digits. Dotted dark red line indicates the ~ 400 N cadaveric ultimate tensile benchmark (Lin et al., 1990; Schöffl et al., 2009).

4.4 Energetic Grip Frontiers & Optimal Grip Transitions

Evaluating the minimum-effort frontier across hold depths reveals clear biomechanical transition boundaries: 1. $d_{hold} < 8.0$ mm (**Micro-Edge Zone**): The **Half-Crimp** is mechanically optimal. It generates sufficient DIP flexion torque without incurring the severe metabolic and extensor co-contraction penalties of the Full Crimp. 2. $8.0 \text{ mm} \leq d_{hold} \leq 18.0$ mm (**Medium Edge Zone**): **Open Hand** and **Half-Crimp** converge in total tendon demand, with Open Hand minimizing A2 pulley strain. 3. $d_{hold} > 18.0$ mm (**Deep Hold Zone**): **Open Hand** provides clear mechanical and energetic advantages. MP engagement eliminates DIP moment requirements, dropping total tendon force below 400 N and minimizing A2 pulley stress.

5. Discussion

5.1 Biomechanical Etiology of A2 and A4 Pulley Ruptures

The clinical literature consistently reports that the A2 pulley is the most frequently injured structure in climbing, followed by the A4 pulley and combined A2/A3/A4 tears (Schöffl et al., 2003; Lutter et al., 2020). Our spatial model highlights two prominent physical mechanisms explaining this vulnerability: 1. **Moment Arm Amplification in Full Crimp**: When the PIP joint flexes to 90° – 100° and the DIP hyperextends, the tendon deflection angle across the distal flexor sheath reaches its anatomical maximum. The resulting bowstringing force vector drives A2 pulley loading to 281.6–299.7 N during standard 100 N hangs, operating within proximity of cadaveric reference rupture benchmarks (~ 400 N; Lin et al., 1990; Schöffl et al., 2009). Under dynamic shock loading (foot slips, 171.7 N), loads spike to 483.6–561.0 N, causing acute traumatic rupture. 2. **Out-of-Plane Shearing**: While planar models assess only normal tensile stress, our 3D formulation demonstrates that MCP radial abduction produces lateral shear forces exceeding 60 N on A2 and 46 N on A4. Because flexor pulleys are anisotropic fibrocartilaginous sheaths optimized for longitudinal hoop stress, transverse shearing causes severe stress concentrations at the distal and lateral margins, initiating microscopic tears that can progress to complete rupture.

5.2 The Anthropometric Dilemma: Phenotypic Advantage vs Mechanical Penalty

A long-standing debate in sports science centers on whether finger length correlates with climbing performance. Our findings indicate that on small edges, **longer digits incur a notable mechanical penalty if internal tendon moment arms do not scale proportionally**: - Under unscaled lever arm conditions, longer phalanges linearly increase the external moment lever arms ($r \times F$) at the DIP and PIP joints without an offset in tendon mechanical advantage. - Consequently, long-fingered athletes are estimated to require up to +15.6% to +16.0% more tendon tension than standard-fingered peers, and up to +37.2%–+38.1% more than short-fingered peers on identical edge depths. - In clinical practice, this helps explain why long-fingered climbers experience a higher incidence of chronic A2 pulley tenosynovitis and why they instinctively adopt open-hand grips on holds where shorter-fingered peers half-crimp comfortably.

5.3 Limitations and Future Research

In accordance with Occam's razor, our framework prioritized physically measurable parameters. However, certain simplifications warrant future extension: - **Quasistatic vs. Dynamic Eccentric Loading**: The current model evaluates static equilibrium. Dynamic movement analysis during foot slips indicates that eccentric uncurling introduces transient inertial and viscoelastic load spikes exceeding static values by 150–200% (Schweizer, 2001; Schöffl et al., 2003). - **Inter-Digit Quadriga Coupling**: Middle and ring fingers share a common FDP muscle belly. Asymmetrical single-finger loading in pocket holds generates high inter-tendinous shear across the lumbrical muscles that warrants multi-digit extension.

6. Practical, Athletic, and Clinical Implications

The biomechanical insights generated by our three-dimensional model provide a physical foundation for evidence-informed climbing coaching, hangboard training protocols, and clinical injury prevention:

6.1 Grip Hygiene and Pulley Load Budgeting

- **The "Pulley-Protective" Open Hand**: Our model demonstrates that transitioning from Full Crimp to Open Hand reduces A2 pulley normal force from 483.6 N down to 57.1 N on a 10 mm edge (an over 85% reduction in fibrocartilage stress; below 100 N across typical athletic loads). In high-volume training (mileage, endurance laps, foundational hangboard blocks), climbers are encouraged to favor the Open Hand or relaxed Half-Crimp.
- **Rationing the Full Crimp**: The Full Crimp introduces passive DIP hyperextension, forcing the FDP tendon to sustain up to 65% of the total muscular load while requiring extensor (EDC) co-contraction to stabilize the terminal phalanx. Full Crimping is best treated as a high-intensity reserve, utilized primarily for sub-8 mm project attempts where DIP flexion torque cannot be achieved through alternative grips.

6.2 Alignment and the Elimination of Transverse Shearing (Side-Pulls & Gastons)

- **Hoop Stress vs. Transverse Shear**: Annular pulleys are anisotropic fibrocartilaginous bands evolved to resist circumferential hoop tension. Our 3D kinematics reveal that a modest 15° MCP radial abduction introduces 61.0 N of transverse lateral shear on A2, 46.4 N on A4, and 44.4 N of mediolateral shear on the PIP collateral ligaments.

- **Forearm-Hold Co-Linearity and Center-of-Gravity (CoG) Optimization:** When engaging side-pulls, gastons, or angled crimps, athletes benefit from dynamically positioning their center of gravity (hips, torso, and foot placements) so that the forearm pulling vector aligns as perpendicularly as possible to the hold edge. Aligning the pull vector optimizes mechanical leverage, reducing the muscular finger effort required to stick the move, while minimizing parasitic out-of-plane torque and transverse pulley shear. Technique that maintains alignment can serve as functional pulley protection. Dropping the elbow or internally rotating the shoulder while bearing down can twist the flexor sheath, increasing edge-peel stresses that contribute to micro-tears.
- **Hangboard and Force Gauge (Tindeq) Training Pitfall:** During hangboard training or isometric dynamometer strength assessments (e.g., using Tindeq Progressor or crane scales), athletes often subconsciously roll their wrists, flare their elbows, or torque digits to record higher peak force readings. This false optimization artificially inflates recorded metrics through wedged joint mechanics and shoulder momentum, but introduces hazardous rotational shear across the A2/A4 sheaths and collateral ligaments. Athletes and coaches are encouraged to prioritize perpendicular alignment, training efficacy, and connective tissue longevity over vanity peak force values.

6.3 Phenotype-Specific Periodization: The "Long-Finger Protocol"

- **The Mechanical Reality of Longer Phalanges:** Under unscaled lever arm conditions, longer digits suffer a +15.6%– + 16.0% force penalty over standard digits (and +37.2%– + 38.1% over short digits) on identical edge depths. Long-fingered athletes operate with an intrinsically narrower margin of structural safety; in Half-Crimp under standard 100 N hangs, unscaled long digits sustain 346.6 N on A2 (compared to 252.7 N for short digits).
- **Prolonged Connective Tissue Periodization:** While skeletal muscle adapts to training stimuli within weeks, collagen synthesis, cross-linking, and pulley sheath remodeling require 12 to 24 months of progressive overload. Long-fingered climbers are advised to adopt conservative, progressive periodization models rather than rapid, aggressive hangboard cycles.
- **Volume and Recovery Management:** Because long digits accumulate tissue micro-strain at elevated rates when lever arms are unscaled, long-fingered climbers typically benefit from extended recovery windows (48–72 hours between high-intensity crimping sessions) and moderate total weekly crimp volume.
- **Tactical and Stylistic Optimization:** Long-fingered climbers should strategically orient their climbing toward slopers, large pinches, open-hand compression volumes, and technical body-positioning problems where span and contact surface area provide a mechanical advantage. On small micro-edges, they should prioritize high-step footwork and drop-knees to sink their weight, engaging the middle phalanx ($d > L_{DP}$) rather than over-relying on high-stress closed crimps.

6.4 Footwork as "Pulley Armor": Mitigating Dynamic Shock Loading

- **The Etiology of Acute Structural Failure:** In static equilibrium on a 10 mm edge, healthy standard digits sustain 281.6–299.7 N of A2 force under a 100 N tip load, operating within physiological limits. However, when a foot unexpectedly slips or blows off a foothold, the external load instantaneously spikes to 25–40% body weight per digit ($F_{tip} \rightarrow 171.7$ N).
- **The Dynamic Load Surge:** At 171.7 N per digit, A2 pulley loads spike to 483.6 N (standard) and 561.0 N (long unscaled), substantially exceeding the ~ 400 N cadaveric ultimate tensile strength and heightening the risk of acute pulley rupture. Precise footwork, core tension, and a conditioned response to release hand load when feet slip are key practical safeguards against shock blowouts.

6.5 Inter-Digit Asymmetry, Ergonomic Rungs, and Quadriga Management

- **The Middle-Finger Overhang:** On conventional flat hangboard rungs, the anatomical length disparity between the middle finger (Digit III) and its adjacent neighbors (Digits II and IV) forces Digit III into hyper-flexion ($> 105^\circ$ PIP flexion) to establish flush contact on the edge. This concentrates disproportionate normal forces and lateral torque onto the middle finger's A2 pulley.
- **Ergonomic Rung Geometry:** Training boards should incorporate anatomical curvature (subtle arc-shaped or offset rungs) that match individual finger length gradients, equalizing contact pressure across digits II–V.
- **Pocket Training and Lumbrical Shear:** Due to the shared deep flexor muscle belly (Quadriga effect), pocket grips that forcefully curl the middle and ring fingers while dropping adjacent fingers into extreme extension produce high inter-tendinous lumbrical shear. Climbers training two-finger pockets are advised to avoid curling dropped fingers into the palm.

6.6 Joint Capsule and Collateral Ligament Trauma: The "Escaping Middle-Finger" Hand-Slip Mechanism

- **The Sequential Disengagement Cascade:** When a hand unexpectedly slips off a hold during dynamic loading, the digits do not leave the edge simultaneously. Because the middle finger (Digit III) protrudes 8–15 mm distally beyond the adjacent index (II) and ring (IV) fingers, Digits II and IV disengage first as the palm moves away from the wall.
- **Dynamic Impulse Concentration on a Single Ray:** For a critical 20–60 ms window, 100% of the escaping body momentum and reaction force is concentrated onto the single protruding middle finger. As the digit is dragged over the hold's crest, it is subjected to violent, high-velocity eccentric extension coupled with out-of-plane torsional shear ($\vec{M}_{rot} = \vec{r} \times \vec{F}_{slip}$).
- **Capsular Pathomechanics in Long Digits:** While annular pulleys primarily fail under circumferential hoop tension (flexor bowstringing), this dynamic shock directly strains the **PIP joint fibrous capsule, collateral ligament complex, and fibrocartilaginous volar (palmar) plate**. In long-fin-

gered phenotypes, the extended moment arms ($L_{MP} + L_{DP}$) dramatically amplify the external torsional torque acting on the PIP joint. This produces micro-tears in collateral ligament origins and capsular synovitis, manifesting clinically as chronic circumferential joint swelling, stiffness, and persistent lateral joint line ache following hand slips.

- **Preventative and Clinical Mitigation:** Climbers with pronounced middle-finger length disparities are advised to resist desperate single-finger clawing saves when slipping, utilize supportive PIP cross-taping (X-taping) or buddy-taping (Digits III and IV) during dynamic bouldering sessions, and treat post-slip capsulitis with active mid-range isometrics rather than aggressive passive joint extension.

6.7 Edge Normalization in Athletic Assessment

- **Eliminating the 20 mm Bias:** Standardized testing protocols routinely utilize an arbitrary 20 mm edge for all athletes. Biomechanically, a 20 mm edge represents a deep hold for an extreme short-fingered climber ($L_{DP} = 18.7$ mm, $d/L_{DP} = 1.07$, transferring load to the middle phalanx and A3 pulley), but a shallow hold for an extreme long-fingered climber ($L_{DP} = 25.3$ mm, $d/L_{DP} = 0.79$, concentrating all force on the distal phalanx). While these $\pm 15\%$ phenotypes represent extreme anatomical boundary cases, even subtle natural variances between 20 mm and 24 mm phalanx lengths substantially skew athletic evaluations. Edge depth in scientific testing and training prescriptions should be normalized to individual anatomy ($d_{hold} = 0.8L_{DP}$).

6.8 Hold Tribology, Skin Care, and the "Dry-Fire" Hazard

- **Normal-Force Friction Decay:** Skin friction follows a non-linear power-law decay ($\mu \propto F_N^{n-1}$). On polished or glassy holds, the drop in friction coefficient forces athletes to generate excessive normal squeezing force, multiplying tendon tension and pulley hoop stresses.
- **The Dual Hazards of Moisture Extremes (Grease vs. Dry-Fire):** Epidermal skin is viscoelastic; friction exhibits an inverted U-shaped relationship with ambient humidity and temperature:
- *The Warm/Humid Extreme:* When holds or skin warm up, a liquid sweat layer acts as a lubricant, precipitating gradual slipping.
- *The Freezing/Arid Extreme ("Dry-Firing"):* Conversely, under extreme cold ($< 5^\circ\text{C}$), very low relative humidity, or chalk-caked conditions, the stratum corneum loses its viscoelastic compliance, becoming rigid and glassy. Without microscopic deformation over rock asperities, the fingertips skate off holds instantaneously with zero tactile warning—the athletic "dry-fire". This instantaneous load release causes violent shock loading on the remaining digits and dynamic joint capsule trauma.
- **Micro-Climate and Skin Optimization:** Empirical observations from climbing tribology literature indicate optimal friction is achieved in cool conditions ($8\text{--}15^\circ\text{C}$, $40\text{--}60\%$ RH). On arid, freezing days, athletes can help prevent dry-firing by avoiding excessive chalk over-caking and gently warming fingertips prior to hard attempts to maintain epidermal compliance. Regular hold brushing removes chalk glazing to preserve rock micro-texture.

6.9 Lifespan Biomechanics: Ontogenetic Phalanx Elongation and Vulnerability Profiles

- **The Ontogenetic Lever Dilemma:** Phalanx segment lengths evolve across the human lifespan, transforming how external grip forces translate into internal tendon tension ($T_{FDP} \approx M_{PIP}/h_{FDP}$) and joint reactions:
- *Pediatric Hold Depth Inversion ($< 12\text{--}13$ years):* Miniature distal phalanges ($L_{DP} \approx 13\text{--}16$ mm) transform standard adult holds ($18\text{--}20$ mm) into deep multi-phalanx ledges ($d/L_{DP} > 1.2$), naturally engaging the middle phalanx and suppressing external DIP moment arms. However, on youth micro-edges ($d < L_{DP}$), single-phalanx tip loading concentrates bending moments across the open cartilaginous physis at the base of the middle phalanx.
- *Primary Periphyseal Stress Injuries (PPSI) in Pubertal Athletes (13–18 years):* During the pubertal growth spurt (Peak Height Velocity), phalanges elongate rapidly. Sports medicine consensus (Schöffel et al., 2023, 2025) demonstrates that **acute annular pulley tears are comparatively rare in children and adolescents**; instead, the open epiphyseal growth plate is structurally the weakest link in the kinetic chain. Repetitive high-load crimping subjects the middle phalanx base to dorsal tensile and shear stresses, precipitating Primary Periphyseal Stress Injuries (PPSI; Stages I through IV, progressing from subclinical bone marrow edema to stress fractures and potential growth arrest). Preventive management mandates avoiding weighted hangboard training and limiting repetitive full crimping during active growth spurts.
- *Adult Phenotypic Divergence (18–45 years):* Skeletal maturity locks phalanx dimensions, where unscaled morphological lever variance produces a $+15.6\%$ penalty (Long vs Standard) and $+37.5\%$ penalty (Long vs Short), motivating phenotype-specific periodization and individualized edge calibration ($0.8L_{DP}$).
- *Masters Articular Cartilage Degradation (45+ years):* Because joint compressive forces scale directly with internal tendon tension ($\vec{F}_{joint} \approx \vec{T}_{FDP} + \vec{T}_{FDS} + \vec{F}_{ext}$), long-fingered climbers accumulate higher cumulative joint contact stress over athletic careers. In aging athletes with reduced chondrocyte turnover, cumulative mechanical work accelerates degenerative joint remodeling and PIP osteoarthritis, encouraging strategic lever-arm de-amplification via open-hand postures and ergonomic hold geometries.

7. Conclusion

We have established a comprehensive three-dimensional musculoskeletal model of the human finger that unifies multi-segment contact mechanics, CT-calibrated spatial moment arms, biological EMG constraints, and out-of-plane pulley shear. The model resolves longstanding anomalies in planar biomechanics, demonstrates the profound impact of hold depth on flexor recruitment, and quantifies the mechanical penalty imposed on longer digits. Furthermore, our translation of 3D spatial mechanics into practical training guidelines bridges the gap between theoretical musculoskeletal physics and on-the-wall athletic longevity. This framework provides an open, reproducible foundation for clinical injury prevention, surgical pulley reconstruction, and evidence-based athletic training in sport climbing.

Declarations

Ethical Approval: Validation utilized publicly available, anonymized cadaveric datasets from Synek et al. (2019, *PeerJ* 7470). No human or animal subjects were directly experimented upon.

Competing Interests: The author declares that he has no competing financial or non-financial interests.

Data & Code Availability: The complete simulation engine, validation scripts, and figure generation routines are open-source and publicly available at GitHub: <https://github.com/igorcerovsky/finger>.

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